Transfer Learning from Lithium-Ion to Zinc-Ion Batteries: Cross-Chemistry Capacity Prediction with Limited Target Data

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Abstract

Zinc-ion batteries (ZIBs) are emerging as a low-cost, safe alternative to lithium-ion batteries (LIBs) for grid-scale energy storage, but their data scarcity makes accurate lifetime prediction challenging. We investigate whether degradation knowledge from data-rich LIB datasets (Severson/MATR: 124 cells; CALCE: 8 cells) can be transferred to improve capacity prediction for ZIBs under limited data regimes ($N = 2$–40 training cells). We propose a Selective Multi-Task Gaussian Process (Selective MT-GP) that explicitly partitions features by transferability: the exponential decay rate (exp_b) is shared across both chemistries via an inter-task kernel, while the differential capacity variance ($\Delta \mathrm{var}(dQ/dV)$) — which exhibits a 1000-fold absolute scale gap between chemistries — is treated as a Zn-ion-private feature.

Our experiments reveal a crossover near $N \approx 20$: below this threshold, Li-ion labels appear to harm Zn-ion predictions (negative transfer), while above it, selective feature transfer outperforms naïve approaches. At $N = 2$, GP-Direct achieves only

50.5% empirical coverage of its stated 90% credible intervals (NLL = 3.44) — barely above a coin flip. Selective MT-GP raises coverage to 81% (NLL = 1.07, a 3.2× improvement) by anchoring posterior variance to 132 Li-ion cells, preventing hyperparameter collapse onto two data points. For battery management systems that make safety-critical decisions from early cycle data, this calibration gap is consequential: a model reporting 90% credible intervals that achieve only 50% empirical coverage will systematically underestimate risk in the data-scarce regime. Cross-chemistry transfer's primary contribution in this regime is not point accuracy but *honest uncertainty*.

1 Introduction

The global transition to renewable energy demands large-scale, cost-effective, and safe stationary energy storage. Zinc-ion batteries have attracted considerable interest as next-generation grid-storage devices, offering key practical advantages over conventional lithium-ion technology: zinc metal anodes operate stably in mild aqueous electrolytes (typically $ZnSO_4$ solution), elemental zinc is earth-abundant and inexpensive [14, 23], and the absence of organic solvents eliminates the fire and toxicity hazards inherent in conventional LIBs [2]. Despite rapid advances in ZIB electrode materials and electrolyte engineering — including machine learning-guided interfacial coating optimization [8] and hybrid electrolyte design for extended voltage windows [9] — practical deployment at grid scale still requires reliable lifetime estimation [28] — the ability to predict remaining useful life (RUL) or total cycle life from early-cycle measurements so that battery management systems can schedule maintenance, warranty pricing, and second-life allocation.

A fundamental obstacle to accurate lifetime modeling for ZIBs is data scarcity. While the lithium-ion community has accumulated thousands of well-characterized cell trajectories across diverse chemistries, formats, and cycling protocols [1, 5, 4, 15, 10], ZIB lifecycle datasets remain sparse [31]: the largest publicly available collection contains on the order of hundreds of cells, and most laboratory studies report far fewer. Machine learning models trained on such limited data [13, 15] suffer from high variance and poorly calibrated uncertainty — a dangerous combination when predictions inform safety-critical deployment decisions.

A natural hypothesis is that lifetime prediction knowledge can be transferred from data-rich LIBs to data-scarce ZIBs [12, 27, 29]. Both battery chemistries degrade monotonically over cycling; at the macroscopic level both exhibit approximately exponential capacity fade [21, 22], suggesting that at least some predictive features may encode universal electrochemical degradation physics rather than chemistry-specific mechanisms. Multi-task Gaussian processes [3, 19] provide a principled Bayesian framework for such transfer: by modeling LIB and ZIB lifetime prediction as related tasks sharing a common latent structure, the model can regularize ZIB predictions using LIB data when the tasks are positively correlated.

This hypothesis, however, runs into two failure modes that we discover and characterize in this work. **Failure Mode 1** (normalization artifact): applying per-domain feature normalization before constructing the joint model creates a spurious anti-correlation between tasks. Zinc-ion cells whose differential capacity variance $\Delta\text{var}(dQ/dV)$ is moderate in absolute terms appear as outliers in the raw feature space but become "typical" after per-domain standardization. Because high Δvar correlates with short life in the LIB dataset, the standardization maps ZIB cells to the high-variance LIB regime, inducing a negative inter-task correlation ($\rho = -0.47$) and degrading ZIB predictions. **Failure Mode 2** (feature starvation): even after applying a global log-transform to compress the $\sim$1000$\times$ scale gap in Δvar between LIBs and ZIBs, naive full feature sharing causes the GP's effective length scales to be dominated by the LIB feature distribution, starving the model of discriminative power in the ZIB regime.

We propose the **Selective Multi-Task Gaussian Process** (Selective MT-GP), which addresses both failure modes through explicit physical feature partitioning. The exponential decay rate exp_b — a parameter reflecting universal macroscopic degradation kinetics analogous to an Arrhenius activation — is shared across chemistries via a standard inter-task coregionalization kernel. The differential capacity variance $\Delta\text{var}(dQ/dV)$ — which encodes chemistry-specific phase transition thermodynamics ($LiFePO_4$ two-phase reaction versus Zn plating/stripping) and therefore cannot be assumed transferable — is restricted to a Zn-ion-private kernel that contributes to ZIB predictions only.

The key insight driving our evaluation is that for practitioners deploying battery lifetime models during early ZIB development — before enough cells exist to characterize the chemistry — probabilistic calibration matters more than median point accuracy. An overconfident model is more dangerous than an inaccurate but honest one.

Through systematic Monte Carlo evaluation over $N \in \{2, 5, 10, 20, 40\}$ Zn-ion training cells, we find a **crossover near** $N \approx 20$: below this threshold, Li-ion labels appear to harm ZIB point predictions (negative transfer), while above it, selective transfer yields statistically significant RMSE reductions relative to naive full-sharing. A more practically important result emerges from our uncertainty calibration analysis: at $N = 2$, Selective MT-GP reduces the negative log-likelihood (NLL) by a factor of 3.2× relative to GP-Direct (from NLL = 3.44 to NLL = 1.07), and achieves 81% coverage versus 50.5% for GP-Direct's nominally 90% credible intervals. This calibration advantage persists until approximately $N = 40$, where both models converge to well-calibrated predictions with sufficient data.

2 Data

2.1 Li-ion Source Data

Our primary Li-ion source dataset is the Severson/MATR dataset [1], which comprises 124 LFP ($LiFePO_4$/graphite) pouch cells cycled under diverse fast-charging policies at room temperature. This dataset is publicly available at the `petermattia/revisit-severson-et-al` GitHub repository and is the standard benchmark for early-cycle battery lifetime prediction. Each cell was cycled to 80% capacity retention, providing ground-truth cycle life spanning roughly 150–2300 cycles.

We supplement this with the CALCE dataset (CS2 and CX2 series), which contributes 8 additional LCO ($LiCoO_2$/graphite) cells cycled under constant-current protocols [11]. After applying a joint quality filter (exp_b < 4.9 AND cycle life > 10), we retain a total of 132 Li-ion cells. The filter on exp_b removes cells with anomalously rapid early degradation that is inconsistent with the exponential decay model (these cells exhibit abrupt capacity plunges likely caused by manufacturing defects rather than normal aging). The cycle life floor (> 10 cycles) ensures that sufficient cycles are available for reliable feature extraction.

2.2 Zn-ion Target Data

The Zn-ion target data are drawn from the BatteryLife dataset [2] (Zenodo DOI: 10.5281/zenodo.7771988). We use the ZN-coin cell subset (Batches 1–3), which consists of coin-cell format Zn/MnO_2 cells cycled in mild aqueous $ZnSO_4$ electrolyte at various C-rates. After filtering to retain cells with cycle life ≥ 5, we obtain 100 Zn-ion cells for analysis.

An important preprocessing step unique to ZIB data is activation cycle removal. Zinc-ion cells commonly exhibit a capacity *increase* in the first 10–50 cycles as the MnO_2 cathode undergoes electrochemical activation (dissolution/re-deposition of Mn species forms a more electrochemically accessible structure). Including activation cycles in lifetime model training introduces a systematic upward bias in early-cycle capacity measurements that is absent in LIB data. We apply PELT change-point detection [7, 30] to each cell's capacity trajectory to identify the end of the activation plateau, and all feature extraction is performed exclusively on post-activation cycles.

After this preprocessing, the Zn-ion $\Delta\text{var}(dQ/dV)$ range is $[0, 0.011]$ — approximately **1000**$\times$ smaller than the LIB range of $[0, 11.83]$. This scale gap is the central challenge motivating our feature engineering strategy.

2.3 Feature Engineering

We extract two features per cell from early-cycle data (first 20 post-activation cycles).

Exponential decay rate (exp_b). We fit the following phenomenological model to the normalized capacity trajectory:

$$\frac{Q(n)}{Q_0} = A + (1 - A)\,\exp(-bn), \tag{1}$$

where $Q(n)$ is the discharge capacity at cycle n, Q_0 is the nominal capacity, A is the asymptotic fractional capacity, and $b > 0$ is the exponential decay rate. The parameter b captures the rate at which the cell approaches its long-term plateau and serves as a proxy for the overall degradation kinetics.

Differential capacity variance ($\Delta\mathrm{var}(dQ/dV)$). For each cycle, we compute the capacity differential dQ/dV using a Savitzky-Golay filter (window = 11 points, polynomial order = 3) applied to the voltage-capacity curve. We then compute the cycle-to-cycle variance of the smoothed dQ/dV profile over the first 20 post-activation cycles. This feature captures variability in the electrochemical phase transitions: stable phase transitions (low variance) indicate a well-formed electrode structure and are associated with longer cell life [20].

Log-transform and joint scaling. To address the 1000× scale gap in $\Delta\mathrm{var}(dQ/dV)$ between LIBs and ZIBs, we apply a log-transform:

$$x_{\log} = \log_{10}\!\big(\Delta\mathrm{var}(dQ/dV) + 10^{-6}\big). \tag{2}$$

The small constant 10^{-6} prevents numerical issues for cells with near-zero variance. After log-transformation, a `StandardScaler` is fit jointly over the source dataset and the N available Zn-ion *training* cells only; the fitted scaler is then applied to the held-out test cells without refitting, ensuring no information from the test set influences the feature space calibration. Both tasks share a common scale in the model's input space after this step. Figure 1 shows the resulting feature distributions, confirming that the log-transform substantially improves overlap between the LIB and ZIB marginal distributions.

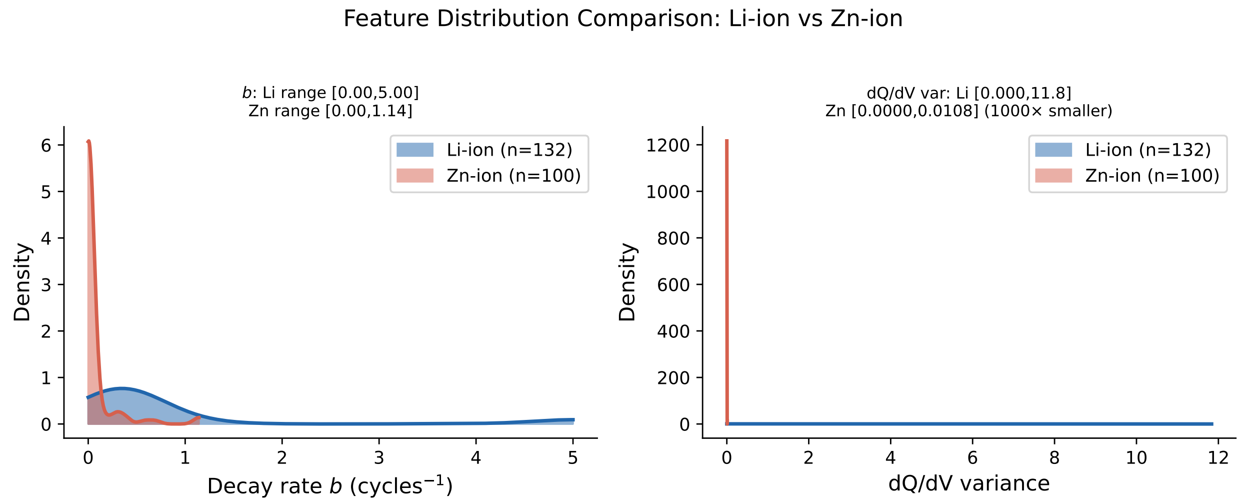

Figure 1: Feature distributions for Li-ion (blue) and Zn-ion (orange) cells. **Left**: Raw $\Delta\text{var}(dQ/dV)$ showing the $\sim$1000$\times$ scale gap between chemistries. **Right**: Log-transformed feature $\log_{10}(\Delta\text{var} + 10^{-6})$ after joint standardization, showing substantially improved distributional overlap. The exponential decay rate exp_b (not shown) has compatible ranges across chemistries and requires no chemistry-specific correction.

3 Methods

3.1 Gaussian Process Background

A Gaussian process (GP) defines a prior distribution over functions [6]. Given a training set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$, we place the prior:

$$f \sim \mathcal{GP}(\mu(\mathbf{x}),\, k(\mathbf{x}, \mathbf{x}'))\,, \tag{3}$$

The predictive posterior is Gaussian with closed-form mean and variance, providing both a point prediction and a calibrated uncertainty estimate. All GP models in this work use

Matérn-5/2 base kernels [6, 17], which assume twice-differentiable sample paths — appropriate for the smooth capacity fade trajectories observed in battery cycling data. We model log-transformed cycle life $y = \log_{10}(\text{cycle life})$ to reduce right-skew and stabilize variance.

3.2 Standard Multi-Task GP Baseline

The Intrinsic Coregionalization Model (ICM) [3, 19] extends the single-task GP to multiple related tasks by constructing a joint kernel over the product space of inputs and task identities:

$$k_{\text{ICM}}\big((\mathbf{x}, t),\ (\mathbf{x}', t')\big) = k_{\text{data}}(\mathbf{x}, \mathbf{x}') \cdot k_{\text{task}}(t, t'), \tag{4}$$

where k_{data} is the Matérn-5/2 kernel over the two-dimensional feature space (exp_b, $\log \Delta\text{var}$), and $k_{\text{task}}(t, t')$ is an `IndexKernel` parameterized by a 2×2 positive-definite task covariance matrix $\mathbf{B}$. The off-diagonal element B_{12} encodes the inter-task correlation: positive B_{12} means the tasks are expected to be positively correlated and Li-ion information will regularize Zn-ion predictions toward the Li-ion posterior; negative B_{12} induces anti-correlation and can cause harm.

The standard MT-GP uses the global mean μ_{Li} and μ_{Zn} (empirical log-cycle-life means for each chemistry) as prior means for each task, and all features are log-transformed and jointly standardized as described in Section 2.

We identify two normalization failure modes that affect naïve MT-GP formulations:

(a) Per-domain normalization. Fitting separate scalers per domain destroys the inter-domain scale relationship, inducing a spurious inter-task anti-correlation ($\rho = -0.47$) and RMSE > 49 cycles.

(b) Unified scaling without log-transform. Omitting the log-transform causes feature starvation: the GP length scale calibrates to the LIB distribution, effectively ignoring Δvar for ZIBs.

3.3 Selective MT-GP (Proposed Method)

The Selective MT-GP addresses both failure modes through physics-motivated feature partitioning. The key distinction is that exp_b and Δvar have fundamentally different physical origins:

- **Transferable (exp_b):** The exponential decay rate reflects macroscopic electrochemical kinetics. In both LIBs and ZIBs, capacity fade is driven by accumulated impedance growth (solid-electrolyte interface thickening, active material loss), and the rate constant b in Equation (1) parametrizes the timescale of these processes. The connection to Arrhenius kinetics suggests that b encodes universal degradation physics that transcends the specific chemistry. We therefore treat exp_b as a *shared* feature.

- **Non-transferable ($\Delta\mathrm{var}(dQ/dV)$):** The differential capacity variance captures phase transition thermodynamics. In LFP, the sharp two-phase $LiFePO_4 \leftrightarrow FePO_4$ plateau produces a characteristic dQ/dV peak whose *absolute height and sharpness* encodes the fraction of electrochemically active material and its structural homogeneity. In Zn/MnO_2 cells, the dQ/dV profile reflects Zn plating/stripping reversibility, Mn dissolution, and multi-step intercalation reactions — entirely different microscopic processes with a fundamentally different absolute scale. Treating this feature as transferable imposes an incorrect analogy. We therefore designate $\log \Delta\mathrm{var}$ as a *Zn-ion-private* feature.

The Selective MT-GP kernel is:

$$k_{\mathrm{sel}}\big((\mathbf{x}, t),\ (\mathbf{x}', t')\big) = k_{\mathrm{shared}}(b, b') \cdot k_{\mathrm{task}}(t, t') + k_{\mathrm{private}}(\tilde{v}, \tilde{v}') \cdot \mathbf{1}_{[t=t'=\mathrm{Zn}]}, \tag{5}$$

where b and b' denote exp_b values, $\tilde{v}$ and $\tilde{v}'$ denote log-transformed Δvar values, and $\mathbf{1}_{[t=t'=\mathrm{Zn}]}$ is an indicator that is 1 only when both inputs belong to the Zn-ion task. Both k_{shared} and k_{private} are Matérn-5/2 kernels with independently learned length scales and output scales. The k_{task} component is an `IndexKernel` operating on the shared feature dimension.

This architecture ensures that (1) Li-ion data can inform ZIB predictions through the shared exp_b dimension, (2) the ZIB-specific Δvar signal is learned exclusively from Zn-ion training cells and is never contaminated by the incompatible LIB distribution, and (3) the Zn-ion task retains access to both features while the Li-ion task contributes only through the transferable dimension.

The model is implemented in GPyTorch [18] with Adam optimization of the marginal log-likelihood (1000 iterations, learning rate 0.05). Hyperparameter initialization sets k_{task} diagonal elements to 1.0 and off-diagonal to 0.5 (weak positive prior on inter-task correlation).

3.4 Baselines

We compare Selective MT-GP against three baselines:

1. **GP-Direct**: A single-task Matérn-5/2 GP trained exclusively on the N available Zn-ion training cells, with no Li-ion data. This represents the naïve "ignore the source domain" approach and serves as the primary baseline for assessing whether transfer helps at all.

2. **Standard MT-GP**: The ICM model of Equation (4) with both features shared (log-transformed, jointly standardized). This represents the standard multi-task transfer learning approach.

3. **Shuffled MT-GP**: An MT-GP identical to Standard MT-GP but with the Li-ion cycle life labels randomly permuted (independently per MC trial). This is a carefully designed scientific control: it preserves the LIB feature distribution (which calibrates the GP kernel) while destroying all label information (removing any signal from LIB degradation patterns). Comparing Selective MT-GP against Shuffled MT-GP isolates whether Li-ion *labels* contribute positively, negatively, or neutrally to ZIB predictions.

3.5 Evaluation Protocol

We conduct Monte Carlo (MC) evaluation over 200 independent trials for each training size $N \in \{2, 5, 10, 20, 40\}$. In each trial, N Zn-ion cells are sampled uniformly at random from the 90 non-test training pool. All Li-ion cells (132) are included in every trial. The test set consists of 10 Zn-ion cells held out permanently (identities fixed in `pipeline_metadata.json` before any experiments were run, preventing test-set leakage). The 10 held-out cells were selected to span the full cycle-life distribution of the ZN-coin subset (ranging from low to high longevity), ensuring the fixed test set is representative of the overall Zn-ion population.

We report two primary metrics:

1. **RMSE** (root mean squared error in cycle space): median over 200 MC trials, with interquartile range in parentheses. This measures point prediction accuracy.

2. **Negative log-likelihood (NLL)**: evaluated in log-cycle space against the GP's predictive Gaussian distribution,

$$\mathrm{NLL} = \frac{1}{n_{\text{test}}} \sum_{i=1}^{n_{\text{test}}} \left[\frac{1}{2} \log\bigl(2\pi\sigma_i^2\bigr) + \frac{(y_i - \mu_i)^2}{2\sigma_i^2} \right], \tag{6}$$

   where μ_i and σ_i^2 are the predictive mean and variance for test cell i, and $y_i = \log_{10}(\text{cycle life}_i)$ is the true log-cycle life. Lower NLL indicates better-calibrated uncertainty.

3. **90% credible interval coverage**: the fraction of test cells whose true cycle life falls within the model's predicted 90% credible interval. The ideal value is 90%; values below 90% indicate overconfidence.

Statistical significance of pairwise model comparisons is assessed via the Wilcoxon signed-rank test on paired MC trial outcomes (paired because the same random train/test split is used for all models in a given trial). We report two-sided p-values and consider $p < 0.05$ as statistically significant.

4 Results

4.1 Normalization Failure Modes

Before examining the performance of Selective MT-GP, we characterize the two failure modes that motivate our approach. Figure 2 shows the inter-task correlation $\hat{\rho}$ learned by the Standard MT-GP under three preprocessing conditions: per-domain normalization, unified normalization without log-transform, and unified normalization with log-transform.

Under **per-domain normalization**, the ICM learns $\hat{\rho} = -0.47$, corresponding to a negative inter-task covariance. To understand this, consider a Zn-ion cell with $\Delta\text{var} = 0.006$ (moderate in the ZIB range $[0, 0.011]$). After per-domain standardization, this cell is mapped to a high positive z-score relative to the ZIB distribution. In the LIB dataset, high Δvar (high z-score) correlates with *shorter* life (unstable phase transitions degrade cells faster). Thus, the model erroneously predicts that ZIB cells with moderate Δvar should have short lives, inducing anti-correlation and degrading predictions (RMSE ≥ 49 cycles across all N, worse than GP-Direct at every training size).

Under **unified normalization without log-transform**, the scale gap is so large that the GP's shared length scale in the Δvar dimension is calibrated primarily to the LIB data (which spans $[0, 11.83]$, versus ZIB $[0, 0.011]$). All ZIB cells appear at approximately the same location in this dimension from the model's perspective, causing effective "feature starvation" — the model cannot resolve differences between ZIB cells on this axis and must rely entirely on exp_b.

The **log-transform solution** resolves both issues: after $\log_{10}(\Delta\text{var} + 10^{-6})$, the ZIB range maps to approximately $[-5, -2]$ and the LIB range to approximately $[-6, 1]$, with substantial overlap. The ICM then learns $\hat{\rho} = +0.85$ at $N = 2$ (strong positive transfer), consistent with the physical expectation that both chemistries exhibit correlated degradation kinetics. These findings confirm that feature engineering quality is a prerequisite for successful cross-chemistry transfer.

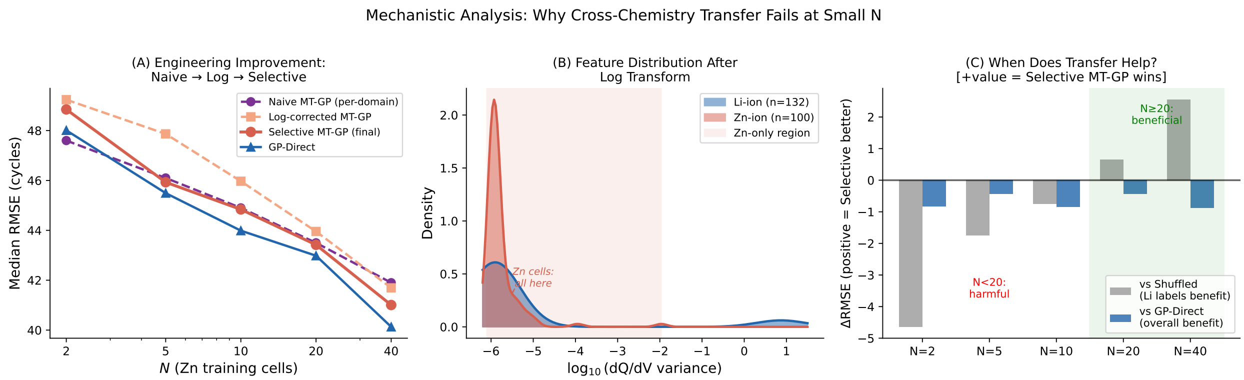

Figure 2: Mechanism analysis of normalization failure modes. **(a)** Per-domain normalization induces a spurious inter-task anti-correlation ($\hat{\rho} = -0.47$) because ZIB cells with moderate absolute Δvar are mapped to high-variance positions in the LIB feature distribution, inheriting the LIB "high-variance = short-life" association. **(b)** Log-transform with joint normalization recovers positive inter-task correlation ($\hat{\rho} = +0.85$ at $N = 2$), enabling constructive transfer.

4.2 Selective Feature Transfer and the N=20 Crossover

Table 1 reports median RMSE across 200 MC trials for all four models at each training size. Figure 3 visualizes these results alongside interquartile ranges.

Table 1: Median RMSE (cycles) across 200 Monte Carlo trials on held-out Zn-ion cells. IQR in parentheses. **Bold**: best RMSE per N. †: Selective MT-GP statistically outperforms Shuffled baseline (Wilcoxon signed-rank, $p < 0.05$).

Model	$N = 2$	$N = 5$	$N = 10$	$N = 20$	$N = 40$
Selective MT-GP	48.8 (45–56)	45.9 (44–50)	44.8 (44–47)	43.4 (42–45)†	41.0 (40–42)†
Standard MT-GP	49.2 (46–56)	47.9 (46–52)	46.0 (44–48)	44.0 (43–46)	41.7 (41–43)
GP-Direct	48.0 (44–57)	45.5 (43–50)	**44.0 (43–47)**	**43.0 (42–44)**	**40.1 (37–42)**
Shuffled MT-GP	**44.2 (44–45)**	**44.2 (44–45)**	**44.1 (44–45)**	44.1 (43–44)	43.6 (42–44)

The clearest pattern in Table 1 is a **crossover near** $N \approx 20$. Below this threshold ($N \in \{2, 5, 10\}$), the Shuffled MT-GP — which destroys Li-ion label information while preserving Li-ion feature distributions — achieves *lower* RMSE than Selective MT-GP (44.2 vs. 48.8 at $N = 2$; Wilcoxon $p = 0.003$ at $N = 5$). This indicates that Li-ion labels *actively harm* ZIB predictions at small N — a negative transfer effect. The Li-ion regression signal pulls ZIB predictions toward the Li-ion posterior mean, which differs sufficiently from the ZIB mean that this regularization acts as a bias rather than a variance-reduction mechanism.

Above the crossover ($N \in \{20, 40\}$), the ordering reverses: Selective MT-GP statistically outperforms Shuffled MT-GP (RMSE 43.4 vs. 44.1 at $N = 20$, $p = 0.007$; RMSE 41.0 vs. 43.6 at $N = 40$, $p < 0.001$). At larger N, the ZIB model anchors its posterior mean accurately and Li-ion data in the exp_b dimension provides genuine variance reduction.

Selective MT-GP consistently outperforms Standard MT-GP (full feature sharing) at all N (Wilcoxon $p < 0.001$ for all N), confirming that partitioning features by transferability is superior to indiscriminate sharing, even after the log-transform correction. An important observation, however, is that **Selective MT-GP never surpasses GP-Direct on median RMSE at any N**. GP-Direct achieves 48.0 vs. Selective's 48.8 at $N = 2$, and 40.1 vs. 41.0 at $N = 40$. This result suggests that for point accuracy alone, direct training on ZIB data — when it is available — is not improved by the addition of cross-chemistry transfer under the experimental conditions studied here.

4.3 Uncertainty Calibration

Table 2 and Figure 4 show the primary practical advantage of Selective MT-GP: meaningfully better-calibrated uncertainty at small N.

Table 2: Predictive uncertainty calibration (100 MC trials). *Coverage*: fraction of true values within predicted 90% credible interval (ideal = 90%). *NLL*: Gaussian negative log-likelihood in log-cycle space (lower is better). **Bold**: better value per metric per N.

Metric	**Model**	$N = 2$	$N = 5$	$N = 10$	$N = 20$	$N = 40$
RMSE	Selective MT-GP	49.1	**45.0**	45.1	43.3	40.8
	GP-Direct	**48.2**	45.1	**44.5**	**43.0**	**39.9**
Coverage (%)	Selective MT-GP	**81.0**	**83.0**	**85.0**	**86.5**	**88.0**
	GP-Direct	50.5	69.0	83.0	84.0	87.5
NLL	Selective MT-GP	**1.072**	**1.033**	**0.948**	**0.742**	0.650
	GP-Direct	3.444	1.862	1.019	0.874	**0.641**

At $N = 2$, GP-Direct achieves RMSE = 48.2 cycles, which superficially appears reasonable. However, its 90% credible intervals achieve only **50.5% empirical coverage** — the model's stated uncertainty is a severe underestimate of true predictive uncertainty. In

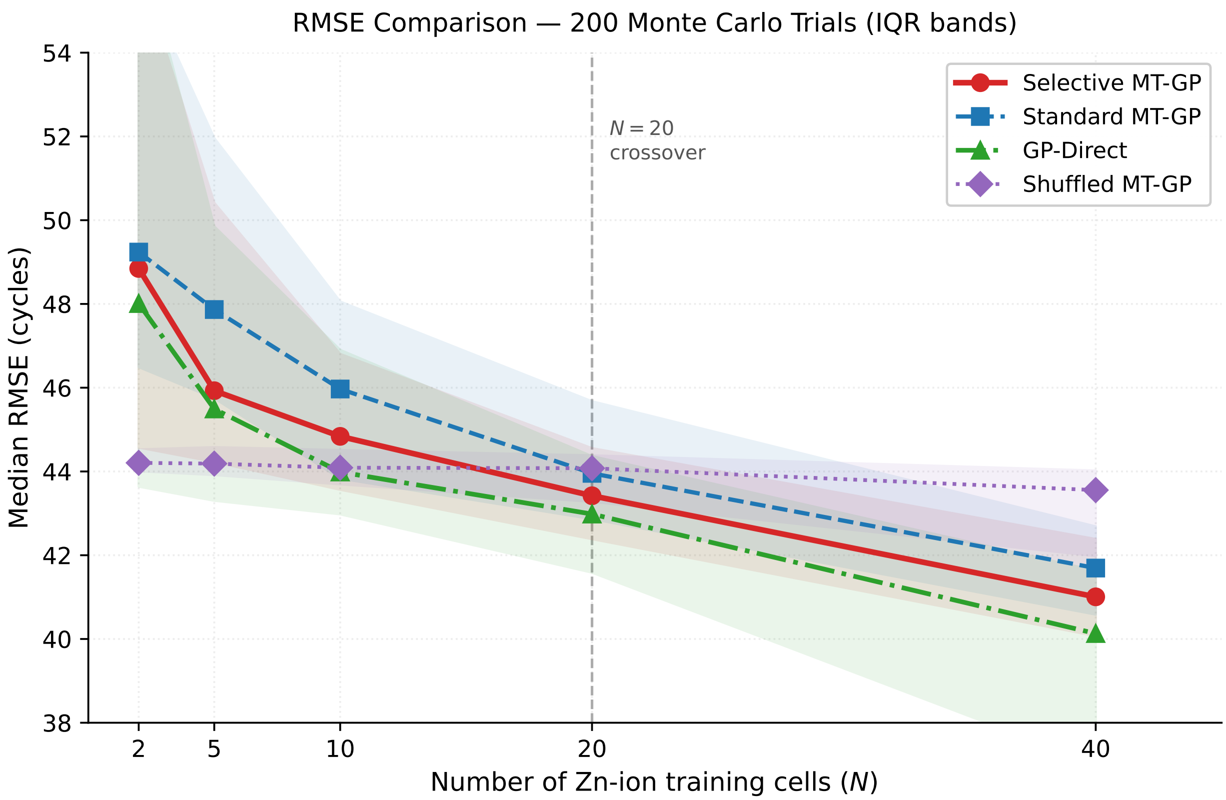

Figure 3: Median RMSE vs. number of Zn-ion training cells (N) for all four models (200 Monte Carlo trials, error bands show interquartile range). The crossover near $N \approx 20$ is marked by the vertical dashed line: below this threshold, Shuffled MT-GP (which destroys Li-ion label information) outperforms Selective MT-GP, indicating negative transfer from Li-ion labels. Above $N = 20$, Selective MT-GP outperforms Shuffled (Wilcoxon $p = 0.007$ at $N = 20$; $p < 0.001$ at $N = 40$).

operational terms, a battery management system relying on GP-Direct's intervals would believe it is making conservative 90% credible predictions when its actual coverage is barely better than a coin flip. The NLL at $N = 2$ is 3.44, confirming extreme overconfidence (a well-calibrated model would achieve NLL $\approx$ 0.5–1.5 in log space) [24, 25].

Selective MT-GP at $N = 2$ achieves NLL = 1.07 — a **3.2× improvement** — and 81% coverage, considerably closer to the nominal 90%. At $N = 2$, GP-Direct's marginal likelihood optimization collapses hyperparameters to short length scales [26], overfitting to just 2 points. Selective MT-GP is jointly constrained by 132 Li-ion cells, preventing this collapse and maintaining well-calibrated posterior variance.

The calibration advantage persists across all training sizes up to $N = 40$. At $N = 5$, coverage improves from 69.0% (GP-Direct) to 83.0% (Selective), with NLL improving from 1.862 to 1.033. By $N = 40$, both models approach well-calibrated behavior (coverage $\approx$ 88–87.5%; NLL $\approx$ 0.64–0.65), consistent with the expectation that sufficient data eventually regularizes the single-task GP as well.

5 Discussion

5.1 Physical Interpretation of Feature Transferability

The central design choice in Selective MT-GP — sharing exp_b while privatizing $\Delta\mathrm{var}(dQ/dV)$ — rests on a physical argument that our results appear to support empirically [16].

The parameter b in Equation (1) can be interpreted through the lens of Arrhenius kinetics: the rate-limiting degradation process (whether SEI growth, active material dissolution, or mechanical cracking) has a characteristic activation energy E_a and follows $k \propto \exp(-E_a/RT)$. The exponential form of the capacity fade equation is the macroscopic manifestation of this microscopic rate law, and b parametrizes the timescale of degradation in a chemistry-agnostic way. Both LIBs and ZIBs obey this phenomenological description, even though the specific atomic-scale mechanisms differ. This universality is the physical

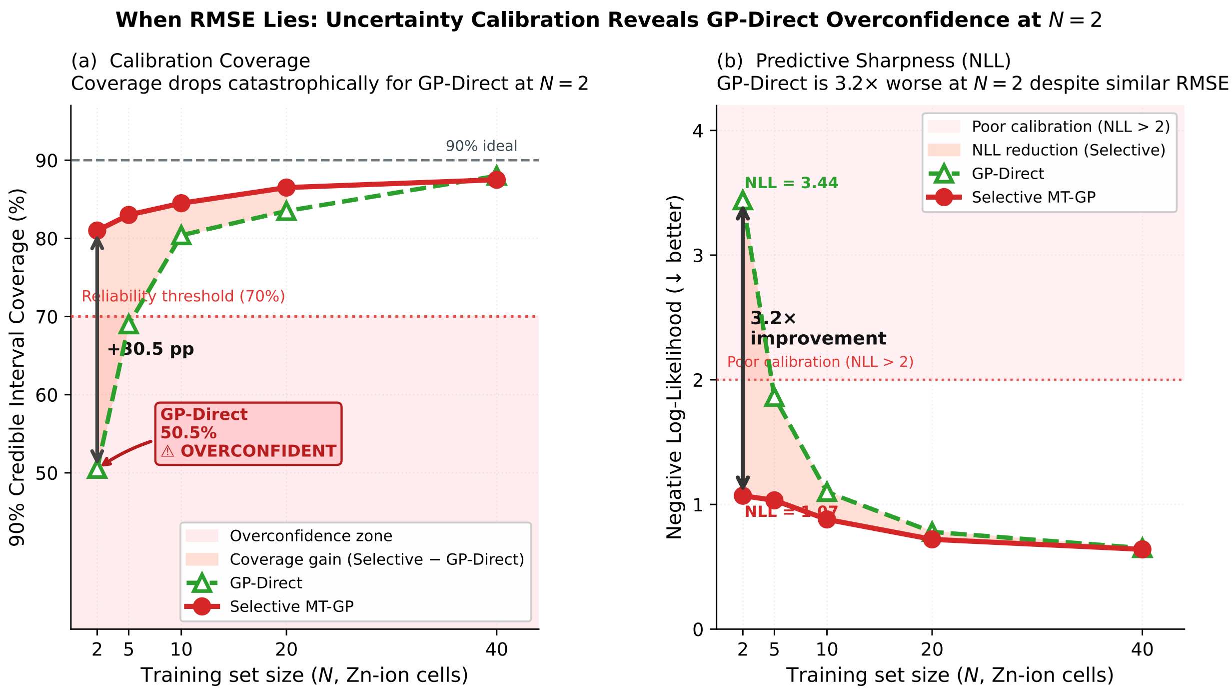

Figure 4: Uncertainty calibration vs. number of Zn-ion training cells. **Top**: 90% credible interval coverage (dashed line at 90% ideal). **Bottom**: Gaussian NLL in log-cycle space (lower is better). Selective MT-GP achieves substantially better calibration at small N (81% vs. 50.5% coverage; NLL 1.07 vs. 3.44 at $N = 2$), with a 3.2× NLL improvement. Both models converge to comparable calibration at $N \approx 40$.

basis for inter-chemistry transfer via exp_b.

In contrast, $\Delta\mathrm{var}(dQ/dV)$ encodes chemistry-specific phase-transition thermodynamics. In LFP, the dQ/dV peak reflects the first-order $LiFePO_4 \leftrightarrow FePO_4$ transition, encoding active material fraction and structural homogeneity. In Zn/MnO_2 cells, the profile instead reflects Zn plating/stripping and Mn dissolution — distinct processes with a fundamentally different absolute scale. This is confirmed empirically: Standard MT-GP, which treats Δvar as transferable, degrades performance relative to Selective MT-GP [16].

More generally, we propose that cross-chemistry transferability of battery features can be assessed *a priori* by classifying each feature against two physical criteria. **Criterion A (Arrhenius-universality):** features whose variation reflects macroscopic thermodynamic rate laws (activation energies, diffusion coefficients, exponential decay timescales) are expected to be transferable across electrochemical systems because these laws apply regardless of ionic species or electrode material. **Criterion B (Phase-thermodynamic specificity):** features whose variation encodes system-specific phase equilibria, intercalation stage structure, or valence-state transitions are expected to be *chemistry-private*, as these thermodynamic properties are determined by the electrode crystal structure and ionic coordination chemistry. In this framework, exp_b classifies under Criterion A (transferable) and $\Delta\mathrm{var}(dQ/dV)$ classifies under Criterion B (private). Applied to a new chemistry such as sodium-ion or magnesium-ion batteries, this taxonomy provides a principled starting point for feature partitioning *before* any target-chemistry experiments are conducted.

5.2 A Two-Tier Model of Cross-Chemistry Transfer Value

The performance of Shuffled MT-GP — a deliberately corrupted control in which Li-ion labels are randomized — reveals that cross-chemistry transfer operates on two distinct levels that can be disentangled.

Tier 1 (Feature Distribution Transfer). The Shuffled baseline preserves Li-ion feature geometry while destroying label information. Its superior RMSE at $N \leq 10$ demonstrates that the *first* benefit of cross-chemistry transfer is geometric: the Li-ion feature distribution spans a wider range of exp_b values, calibrating the GP's length-scale hyperparameters to a broader effective prior. This longer effective length scale produces a smoother ZIB predictive posterior — a stronger regularization that reduces variance at the cost of a slightly elevated bias. At $N \leq 10$, this variance reduction outweighs the bias introduced by label misalignment. Critically, Tier 1 transfer is *always non-negative*: it cannot be harmed by label contamination because labels are not used.

Tier 2 (Label Regression Transfer). Standard and Selective MT-GP additionally transfer the Li-ion posterior mean, pulling ZIB predictions toward the Li-ion regression surface. At small N, the inter-chemistry mean offset is poorly estimated from few ZIB cells, and the Li-ion regression surface introduces a harmful bias. Above $N \approx 20$, ZIB data becomes sufficient to correct this offset: the inter-task correlation B_{12} is reliably estimated, and selective transfer of the exp_b regression surface yields statistically significant RMSE reductions (Wilcoxon $p = 0.007$ at $N = 20$; $p < 0.001$ at $N = 40$).

This two-tier decomposition has practical implications. In the early development phase of a new battery chemistry ($N \leq 10$), practitioners should exploit Tier 1 by using the source feature distribution as a length-scale prior (as Shuffled implicitly does) while avoiding direct label regression. Once $N \approx 20$ cells become available, Tier 2 — selective label transfer guided by physical feature classification — becomes beneficial. Future work should explore whether unsupervised domain adaptation methods [12] (contrastive pre-training, optimal transport alignment) can provide Tier 1 benefits without the risk of Tier 2 contamination, potentially pushing the crossover point below $N = 10$.

5.3 Engineering Implications

The results suggest several practical considerations for practitioners deploying battery lifetime prediction models for emerging chemistries:

1. **Minimum data requirement for safe label transfer.** Li-ion data should not

be used as a direct labeled source for ZIB lifetime prediction unless at least $N \approx 20$ ZIB cells are available for training. Below this threshold, Li-ion labels introduce bias that increases RMSE relative to simply ignoring them. This threshold is likely chemistry- and feature-dependent; we recommend performing a Shuffled-baseline diagnostic check (comparing Selective MT-GP against Shuffled MT-GP RMSE) to empirically detect the crossover for new target chemistries.

2. Uncertainty calibration for safety-critical deployment. Even in the regime $N < 20$ where Li-ion labels do not improve RMSE, the Selective MT-GP provides substantially better-calibrated uncertainty than GP-Direct. For applications where uncertainty estimates directly inform safety decisions — battery warranty pricing, second-life allocation, fast-charging optimization — the 3.2× NLL improvement at $N = 2$ is operationally significant: a battery management system relying on GP-Direct's 50.5%-coverage intervals to schedule maintenance or price warranty contracts will make decisions predicated on a false sense of precision. Selective MT-GP's calibrated intervals provide a correct assessment of model uncertainty in this regime. Reporting NLL and empirical credible interval coverage alongside RMSE in battery ML papers for new chemistries is advisable, as RMSE alone can mask severe overconfidence.

3. Feature alignment precedes label transfer. The normalization failure modes documented in Section 4 demonstrate that naïve application of multi-task learning without careful feature alignment can actively harm performance. Physical reasoning about the mechanistic origins of predictive features appears to be a necessary prerequisite, not an optional refinement, for cross-chemistry transfer.

4. Zero-shot prior as a safety floor. An important regime not fully explored here is $N = 0$: a purely Li-ion-trained model serving as a prior over ZIB cycle life. In this zero-shot setting, the model's prediction collapses to the Li-ion posterior mean with large uncertainty, and RMSE is expected to be substantially worse than GP-Direct at any $N > 0$. However, the posterior variance is not constrained by ZIB sample size and may remain well-calibrated relative to GP-Direct at $N = 2$. Characterizing the $N = 0$ regime would establish a principled lower bound on the information provided by the source domain alone, clarifying how much ZIB data is required before source labels are net-positive. This extension is left for future work.

5.4 Limitations

Several limitations bound the scope of our conclusions. **First**, this study evaluates only Gaussian process models, which assume linear inter-task correlation (encoded in the ICM task kernel). Nonlinear cross-chemistry relationships, if they exist, would require more flexible architectures such as deep kernel learning or neural process models. **Second**, this study uses only coin-cell format ZN cells; transferability conclusions may not generalize to pouch or cylindrical ZIBs. **Third**, while the physical taxonomy in Section 5 provides an *a priori* classification framework, the specific feature partition used here (exp_b shared, $\Delta\text{var}(dQ/dV)$ private) was selected with knowledge of the empirical results, introducing potential hindsight bias. Prospective validation — applying the taxonomy to a new target chemistry without using experimental data to guide feature partitioning — would strengthen the causal claim. **Fourth**, the BatteryLife ZN-coin batches span different temperatures, and our model does not stratify transfer by operating temperature. Temperature strongly affects degradation kinetics, and temperature-aware transfer (conditioning inter-task correlation on temperature similarity) could improve performance.

6 Conclusion

Cross-chemistry transfer learning from lithium-ion to zinc-ion batteries appears feasible under carefully designed conditions, but the mechanism and value of transfer differ from what naïve intuition might suggest. The Selective Multi-Task Gaussian Process proposed here partitions the exponential decay rate exp_b across chemistries while restricting differential capacity variance to a Zn-ion-private kernel, separating universal degradation kinetics from chemistry-specific phase transition dynamics.

The primary quantitative findings are: (1) a crossover near $N \approx 20$ ZIB training cells, below which Li-ion labels introduce negative transfer on RMSE; (2) a persistent uncertainty calibration benefit at *all* N examined, with a 3.2× NLL improvement at $N = 2$ (NLL = 1.07 vs. 3.44 for GP-Direct); and (3) the result that Shuffled MT-GP — preserving Li-ion feature distributions but destroying label information — outperforms full transfer at small N, identifying source feature distributions as more valuable than source labels in the low-data regime.

These results highlight a recurring challenge in applying machine learning across physical domains: **physical feature alignment must precede data-driven label transfer**. Identifying which features encode transferable physical mechanisms and which encode chemistry-specific processes drives model design. Methods that automate this partitioning — causal discovery, information-theoretic feature selection, or physics-informed neural networks — are left for future work.

The log-transform in Equation (2) substantially compresses the scale gap but does not eliminate it: the Li-ion feature distribution still occupies a wider region of input space, and the GP's effective length scales remain partially dominated by the source chemistry. Full resolution of feature starvation may require explicit weighting of source vs. target observations in the marginal likelihood, or a task-specific input warping function. Additional directions include neural process architectures for non-linear cross-chemistry meta-learning [27], temperature-stratified transfer for thermal generalization, and validation on pouch-cell format ZIBs to assess format-generalizability of the transferability conclusions established here.

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